Morning Wake-Up Circuit

Week 7 Assignment - Problem 1 (Basic)

Ringable lets unrelated devices share one operation. AlarmClock and Doorbell implement the interface directly, and ringAll dispatches ring() polymorphically.

Java solution

```
interface Ringable {
    String ring();
}
class AlarmClock implements Ringable {
    private final String time;
    AlarmClock(String time) { this.time = time; }
    public String ring() {
        return "Alarm ringing for " + time;
    }
}
class Doorbell implements Ringable {
    private final String location;
    Doorbell(String location) { this.location = location; }
    public String ring() {
        return "Doorbell ringing at " + location;
    }
}
public class Main {
    static void ringAll(Ringable[] devices) {
        for (Ringable device : devices) {
            System.out.println(device.ring());
        }
    }
    public static void main(String[] args) {
        AlarmClock a = new AlarmClock("7:00 AM");
        Doorbell d = new Doorbell("Front Door");
        ringAll(new Ringable[] { a, d });
    }
}
```

Sample output

```
Alarm ringing for 7:00 AM
Doorbell ringing at Front Door
```

Note: Both classes implement Ringable directly; they do not need a shared parent class.